

ADVANCED COMPUTER GRAPHICS - FINAL EXAM

MODEL ANSWER (2)

Reference Textbook: Computer Graphics: C Version (2nd Edition) by Hearn & Baker

Exam Date: 11/9/2025 | Total Marks: 70 Marks



Question 1: Answer the following (40 Marks)

Q1.1: What are the capabilities of advanced computer graphics?

- **Photorealistic Rendering:** Accurate light, shadows, and reflections.
- **Texture Mapping:** Realistic surface detailing (wood, fabric).
- **Complex Object Modeling:** Advanced surfaces (Splines, Bezier, quadrics).
- **Physical Animation:** Simulating natural elements (water, smoke, fire).

Q1.2: What is rendering?

Rendering is the process of generating a 2D digital pixel image from a 3D scene description (models, lights, textures, and cameras) using specialized software.

Q1.3: Draw figure illustrate Cathode Ray Tube (CRT)?

Diagram Flow: [Heating Filament] → [Cathode] → [Control Grid] → [Focusing Anode] → [Deflection Plates] → [Phosphor-Coated Screen].

Q1.4: Write the operating characteristics for the LCDs and Raster refresh systems.

LCDs: Flat, non-emissive displays that use liquid crystals to block/pass light based on applied voltage.

Raster Refresh: Scans the screen line-by-line; stores pixel data in a **Frame Buffer**.

Q1.5: Explain how VR systems are used in design and list other applications.

In Design: Real-time interaction with 3D prototypes and architectural walkthroughs.

Other: Medical training, flight simulators, and military training.

Q1.6: Draw figure show Basic video-controller refresh operations?

Workflow: [Frame Buffer] → [Video Controller] → [Monitor/CRT].

Q1.7: Write the steps of Bresenham's Algorithm?

1. Input endpoints (x_0, y_0) and (x_n, y_n) .
2. Plot (x_0, y_0) . Calculate $\Delta x, \Delta y$.
3. Initial decision parameter: **$P_0 = 2\Delta y - \Delta x$** .
4. If $P_k < 0$, next point is (x_{k+1}, y_k) and $P_{k+1} = P_k + 2\Delta y$.
5. If $P_k \geq 0$, next point is (x_{k+1}, y_{k+1}) and $P_{k+1} = P_k + 2\Delta y - 2\Delta x$.

Q1.8: What the function CRT?

Projecting a sharp, focused beam of electrons onto a phosphor screen to emit light and display images.

Q1.9: Write the steps of DDA Algorithm?

Calculate Δx , Δy . Steps = $\max(|\Delta x|, |\Delta y|)$. Calculate increments for x and y. Loop and plot rounded values.

Q1.10: DDA Algorithm Example: Line from (1,7) to (11,17).

$\Delta x = 10$, $\Delta y = 10$. Steps = 10. Increment_x = 1, Increment_y = 1.

Step (k)	x	y	Plot (x,y)
Start	1	7	(1, 7)
1	2	8	(2, 8)
...	...	...	...
10	11 guest	17	(11, 17)



Question 2: Complete the following (20 Marks)

1. Operation of most monitors is based on **CRT**.
2. The **Electron Gun** emits a small spot of light.
3. Bresenham's is based on calculating **decision parameters** (or dy/dx increments).
4. Rasterization in scan line order is **scan conversion**.
5. Digital devices active a **connection** (or sequence) of pixels.
6. Equation for plane surface: **$Ax + By + Cz + D = 0$** .
7. OpenGL provide **polygon** functions.
8. Ellipsoidal surface: **$(x/rx)^2 + (y/ry)^2 + (z/rz)^2 = 1$** .
9. Remove structure: **closeStructure** (or deleteStructure).
10. Place structure within another: **executeStructure** (or openStructure).



Question 3: True or False (10 Marks)

1. **[T]** 3D rotation is not same as 2D rotation. (Needs axis of rotation)
2. **[F]** Change size using parallel transformation. (Correct: Use Scaling Transformation)
3. **[T]** Translation is repositioning along a straight-line path.
4. **[F]** In 3D, any two non-parallel lines meet. (Correct: They can be Skew Lines)
5. **[T]** openStructure(id) is used to create a structure.
6. **[T]** Polygon mesh defines polyhedron features precisely.
7. **[F]** Post structure is used to store picture. (Correct: It displays/renders the structure)
8. **[T]** deleteElementRange(k1, k2) deletes elements.
9. **[F]** Set edit mode with getEditMode. (Correct: Use setEditMode)
10. **[T]** Adjacent pixels likely have same characteristics.